Syllabus MAT 341 – Linear Algebra
Fall 2020, Section 01

Instructor: Xiuquan Wang
Virtual Zoom Office Hours: 10:00 – 10:50 AM every Friday
Email: xwang@tougaloo.edu

Text and Resources:

A. Textbook: Strang, Gilbert. Introduction to Linear Algebra. 5th ed. Wellesley-Cambridge Press, 2016. ISBN: 9780980232776
B. Online Materials
   1. Notes in PDF form for printing or to use electronically within Perusall;
   2. Lecture videos are accessed through the modules in Moodle.
   3. Perusall - Perusall is a collaborative annotation platform for textbooks and documents that allows students to ask questions, answer questions posed by classmates, make insightful comments for each other, and interact with classmates and instructors as they're learning the content. you will be required to interact with the notes with classmates through class assignments in Perusall. Please read the Perusall Grading, Perusall Rubric and Perusall Scoring Examples documents located at the link before proceeding. Do not access the notes in Perusall by going to perusal.com, but only access through the link in Moodle.

Course Description:
This is a basic subject on matrix theory and linear algebra. Emphasis is given to topics that will be useful in other disciplines, including systems of equations, vector spaces, determinants, eigenvalues, similarity, and positive definite matrices

Course Objectives:
The goals for MAT341 are using matrices and also understanding them. Here are key computations and some of the ideas behind them:

- Solving Ax = b for square systems by elimination (pivots, multipliers, back substitution, invertibility of A, factorization into A = LU)
- Complete solution to Ax = b (column space containing b, rank of A, nullspace of A and special solutions to Ax = 0 from row reduced R)
- Basis and dimension (bases for the four fundamental subspaces)
- Least squares solutions (closest line by understanding projections)
- Orthogonalization by Gram-Schmidt (factorization into A = QR)
- Properties of determinants (leading to the cofactor formula and the sum over all n! permutations, applications to inv(A) and volume)
- Eigenvalues and eigenvectors (diagonalizing A, computing powers A^k and matrix exponentials to solve difference and differential equations)
- Symmetric matrices and positive definite matrices (real eigenvalues and orthogonal eigenvectors, tests for x'Ax > 0, applications)
- Linear transformations and change of basis (connected to the Singular Value Decomposition - orthonormal bases that diagonalize A)
- Linear algebra in engineering (graphs and networks, Markov matrices, Fourier matrix, Fast Fourier Transform, linear programming)
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Course Format: Online, asynchronous. Though you generally work at your own pace in a given week, there are open and close dates and deadlines for each week's work.

Final Grade Components

Component	Points	Weightage
Perusall Engagements (15)	**10 point each**	**10%**
Homework (10)	**15 point each**	**15%**
Quizzes (12)	**10 point each**	**10%**

Midterm Exams (2)	**100 point each**	**20% each (40% in total)**
Final Exam	**150 points**	**25%**
Total	**600 points**	**100%**

Letter grade will be determined as follows:

Percentage	Letter Grade
88%-100%	**A**
75%-87%	**B**
62%-74%	**C**
50%-61%	**D**
Below 50%	**F**

General Policies and Information

- **Perusall Engagements** - For each section of material there will be a 10-point assignment which will consist of the notes for those sections (100 points in all). In each assignment, you will be required to ask thoughtful questions, answer a classmate's questions, or make a useful comment about a points of clarification (these are referred to as annotations). It is recommended that you enter multiple annotations (see Perusall Rubric in Moodle). All assignments before a given exam will be due the week of the exam, giving plenty of time to interact with the notes and each other before the exam. Please do not wait until the last week to begin the assignment.
- **Homework:** I will assign homework from the sections as we discuss them. Homework must be turned in on Moodle. It is imperative that you keep your studies current. If needed, we will discuss questions over homework that you have attempted at the beginning of each meeting. Be sure that you are prepared to ask questions over material that we discussed in the previous meeting. You should come to speak with me during my virtual Zoom office hours if your questions do not address the section covered in the previous meeting.
- **Quizzes**: Quizzes will be given on the basic definitions and the statements major theorems in Moodle.
- **Exams**: There will be 2 exams during the semester and a cumulative final exam: The exams will be showed in details. I may break a longer problem into smaller steps, and have you give an answer at each step so you can show your work and so I can give you partial credit when appropriate.

You earn your grade by communicating your understanding of the material through the Perusall engagement, homework, the quizzes and tests. Clearly communicating mathematics will be essential in this course.

Extra Credit: You may earn additional points in the following ways:

- COURSE PARTICIPATION (8 points): We encourage you to utilize instructor's virtual office hours and the MAT341 Discussion Board regularly to ask and answer questions about course material and homework. You can earn up to 4 bonus points to participate in virtual office hours and up to additional 4 points by responding to other students' math questions. Answers must be appropriate and relate to course material to earn credit.
- EXAM PREPARATION (2 points): An exam review sheet is posted in Moodle for each test and students can earn up to 2 bonus points by submitting their work one day before the test. In addition, a bonus exam review assessment will be available in virtual office hours before each test and the assessment will include questions from previous MAT341 exams so that you will have a flavor of the type of questions that you will see on the actual test. Detailed instructions for earning extra credits will be posted with the exam reviews.

MAKE-UP POLICIES AND EXTENSIONS:

- EXTENSIONS ON HOMEWORK: While it is a much better strategy to work ahead, occasionally you may fall behind. You can request an extension on homework within 1 day after the deadline and you will have 24 hours to complete it after extension request. The extension must submit in Moodle directly. However, there will be a 20% grade penalty for those problems completed after the original due date for the assignment.

- MAKE-UP QUIZZES: We offer 2 drops on quizzes, so we do not provide make-ups for online quizzes. To allow time for computer issues, it is recommended not to wait until final hours on the due day to start your quizzes.
- MAKE-UP EXAMS: Students must provide valid documentation for requesting a make-up exam due to a scheduling conflict by the end of the fourth week to avoid penalty. If illness or other extenuating circumstances force you to miss an exam, contact me as soon as possible (no later than 24 hours after the exam) for approval to schedule a make-up exam. We do not consider traveling or lack of internet access as a valid excuse for a make-up exam.
- OTHER MAKE-UPS: There are no make-ups on any extra credit assignments.

INCOMPLETE GRADE: A student who has completed a major portion of the course with a passing grade but is unable to complete the final exam or other course requirements due to illness or emergency may be granted an incomplete, indicated by a grade of I. This allows the student to complete the course within the first six weeks of the following semester. The student must contact Dr. Wang before finals week for departmental approval and must provide documentation of the extenuating circumstances preventing him or her from taking the final exam. The grade of "I" is never used to avoid an undesirable grade, and does not allow a student to redo work already graded or to retake the course.

Note to seniors and other students:
If you are taking this course to satisfy the graduation requirements, I cannot treat you any different from any other student. Your final grade is strictly as a result of your performance in homework, quizzes, tests, and other class activities.

ADA ACCOMMODATIONS: Students with disabilities who qualify under the Americans with Disabilities Act (ADA) and require accommodations should be referred to Dr. Melissa McCoy, the ADA Compliance Officer, at (601) 977-7783, or mmccoy@tougaloo.edu. Disabilities covered by ADA may include learning, psychiatric, physical disabilities, or chronic health conditions. The ADA will provide a documentation letter to the student to present to me. This must be done as early as possible in the semester, at least one week before the first exam, so there is adequate time to make proper accommodations and the accommodations are not retroactive.

TITLE IX POLICY ON SEX DISCRIMINATION, SEXUAL MISCONDUCT AND RETALIATION
Title IX of the Educational Amendments of 1972 ("Title IX") explicitly prohibits sex-based discrimination by any institution of higher education that receives federal funds (which includes Tougaloo College). Under the scope of this federal law and related federal and state laws, Tougaloo must adhere to specific regulations regarding how to address reports of sexual discrimination, sexual misconduct, including sexual harassment, sexual assault, sexual exploitation, indecent exposure, intimate-partner violence including dating violence and domestic violence, romantic/sexual consensual relationships between people with unequal power, stalking, and other misconduct that is sex or gender-based.

Title IX also mandates that a recipient may not retaliate against any person for opposing an unlawful educational practice or policy, or made charges, testified or participated in any complaint action under Title IX. For a recipient to retaliate in any way is considered a violation of Title IX.

Tougaloo College is committed to establishing and maintaining a community that is rich in equality and reflects and promotes personal integrity, civility, and mutual respect. Tougaloo seeks to create an environment in which the greatest holistic academic potential of students and professional potential of employees may be realized. In order to create and maintain such an inclusive climate, all who learn, live and work at Tougaloo are partners in the effort of ensuring that the community is free from sex discrimination, sexual misconduct and retaliation.

Inquiries about Title IX as implemented at Tougaloo, or reports of any violation of Title IX should be directed internally to:

Dr. Linda J. Daniels
Office of the President
Blackmon Administration Building
Telephone: 601-977-4462

Email: ldaniels@tougaloo.edu

Inquiries may be made externally to:
Office for Civil Rights (OCR)
U.S. Department of Education
400 Maryland Ave., SW, Washington, DC 20202-1100
Telephone: (800) 421-3481
TDD: (877) 521-2172
Email: OCR@ed.gov
www.ed.gov/ocr